

Real-Time Implementation of Modified LD-NET using Docker on NVIDIA Jetson Nano for Image Dehazing Application

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Abstract— Over the past few years, the majority of the industrial and surveillance sectors have observed visual sensor networks that record various images of the surrounding environment. The total surveillance systems are impacted by contaminated suspended air particles, which lowers the visual quality of the photos in stormy weather. Haze can interfere with motion analysis algorithms, leading to inaccurate tracking, trajectory prediction, or behaviour analysis of moving objects. The reduced visibility caused by hazing makes it difficult to capture and analyse the motion patterns and velocities of objects accurately. To tackle these challenges, we have used the Light-Dehaze Net (LD-Net) based on the Convolutional Neural Network algorithm in Deep Learning. Our suggested LD-Net uses a transformed atmospheric scattering model to jointly estimate the transmission map and the atmospheric light, in contrast to current learning-based methods that measure the transmission map and the atmospheric light separately. We implemented this LD-net model on NVIDIA Jetson Nano for real-time image dehazing application. Furthermore, we have added a Docker container to this implementation, where this container encapsulates the application and all its dependencies, including libraries, frameworks, and runtime environments. This portability allows the containerized application to run consistently across different environments.

Keywords—Haze removal, Convolutional neural network, LD-Net, single image dehazing, Jetson Nano, Docker Container.

I. INTRODUCTION

The modern technological era has accelerated the development of computer vision systems [1], [2], which have been used in a variety of applications such as surveillance [3], object detection [4], segmentation [5], image/video denoising [6], compression [7], crowd behaviour understanding [8], and inpainting [9]. These systems rely on digital cameras to take high-resolution pictures for a range of uses, from activity recognition to vehicle detection. However, during stormy weather, particularly on misty or foggy days, light is primarily impacted by suspended droplets and other particles. As a result, haze in images is caused by the presence of particles, dust, and other atmospheric pollutants that scatter and absorb light as it travels from the scene to the camera sensor. This scattering and absorption of light lead to a loss of contrast, colour distortion, and reduced visibility in the captured images. This phenomenon is commonly known as "atmospheric scattering." Fog is a type of atmospheric haze that occurs

when tiny water droplets or ice crystals are suspended in the air near the ground, reducing visibility and creating a mist-like appearance. The intensity of fog can vary, leading to these: 1) Thin fog refers to a less dense and less obstructive form of fog, visibility may be reduced to some extent, but objects and structures in the immediate vicinity are still relatively visible, 2) Thick fog causes objects and structures even a short distance away may become barely visible or completely obscured, 3) Dense fog causes greatest impact on visibility and it is highly reduced. These can lead to disruptions in transportation, particularly air and road travel. Flights may be delayed or cancelled due to reduced visibility at airports. On roads, this leads to accidents. This scattering effect reduces the amount of light reaching the camera lens, resulting in a hazy or blurry appearance in the captured images and videos. The reduced visibility can make it difficult to identify objects, people, or events captured by the CCTV camera. Our method to tackle this challenge is dehazing the captured image.

Dehazing is a computer vision and image processing technique used to remove or reduce the effects of atmospheric haze from images. It improves the visual quality of hazy images by restoring the details and colours that were obscured by the haze. Two components can be measured to obtain the results of the dehazing image. 1) Global Atmospheric Light and 2) Medium of Transmission. There are several methods and techniques to approach the results of these two components. These are Dark Channel Prior (DCP) [10], Atmospheric Scattering Model [11], Single image dehazing technique [12], Hazy image reconstruction algorithm [13], State-Of-The-Arts (SOTA), Prior-image dehazing and so on. All of these methods, however, estimate the atmospheric light and transmission maps individually, which increases computer complexity, takes too long, causes colour distortions, a hollow effect, lowers image quality, and has dark effects. The Deep Learning-Based Single Image Dehazing approach was put forth by H. Ullah et al. [14] and is a practically tested method for dehazing images. Convolution Neural Network Algorithm and the updated Atmospheric Scattering Model are combined to jointly approximate the atmospheric light and the transmission map. Here, the reconstruction of hazy images uses the Convolutional Neural Network (CNN) architecture known as Light Dehaze Net (LD-Net).

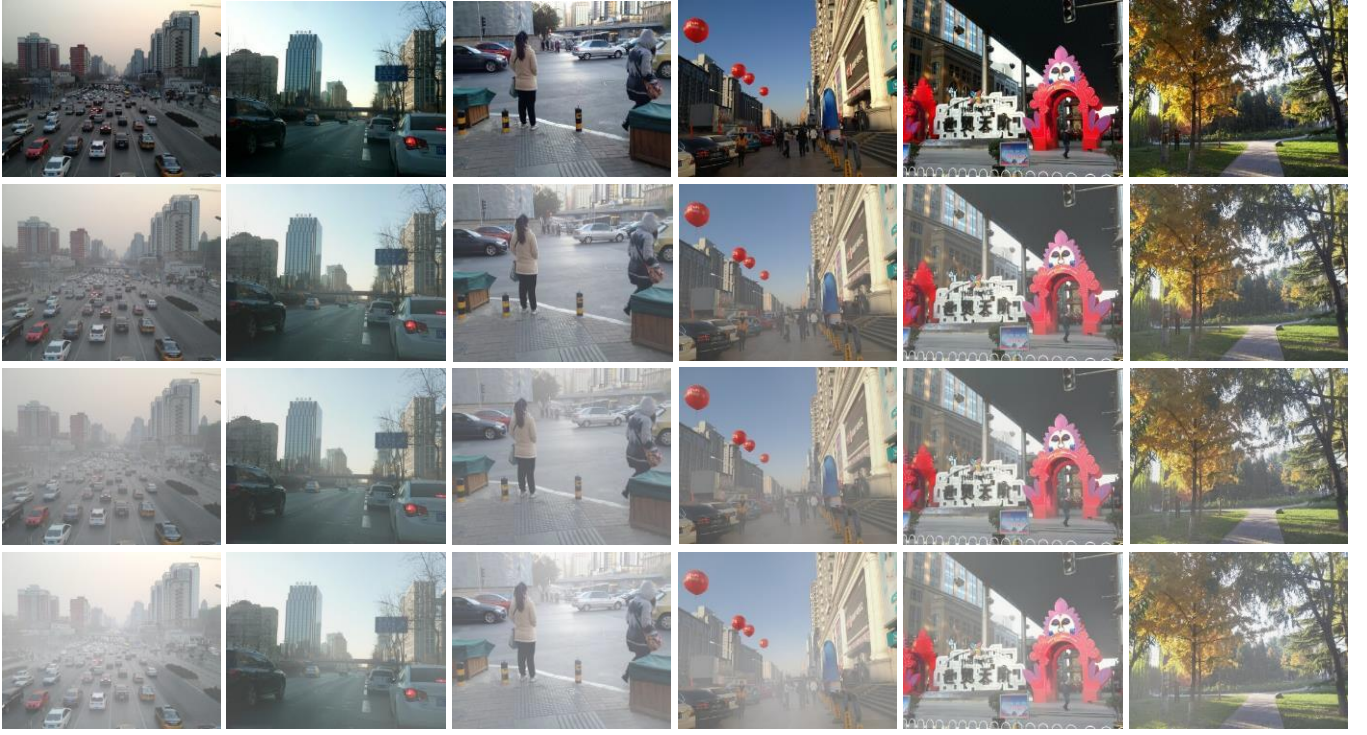


Fig. 1. Sample training images from the dataset. The top row displays photographs without haze, and the next rows display hazy images at varying intensities.

II. PROBLEM FORMULATION

The noise (haze) in a photograph that was taken in a hazy setting typically impairs the visibility of the image. To eliminate the hazy particles from the acquired image, reconstruction of the hazy image is a very time-consuming task. Different image dehazing algorithms are offered to address this issue, but the majority of these techniques suffer from loss of contrast, colour deterioration, and loss of small details. Dehazing algorithms need to effectively recover this lost information without introducing artefacts or over-enhancement. Estimating the atmospheric light (A) accurately is crucial for image dehazing. However, finding the correct atmospheric light value can be challenging, especially in scenes with complex lighting conditions or when the scene contains bright light sources. Estimating the transmission map ($t(x, y)$) is a critical step in dehazing algorithms. However, obtaining an accurate transmission map is difficult due to variations in haze density, complex scene structures, and the presence of noise in hazy images. Some dehazing algorithms, particularly those based on sophisticated physical models or deep learning techniques, can be computationally intensive. Real-time dehazing for high-resolution images or videos can be challenging to achieve.

Many CNN dehazing approaches have very complex architectures that increase the overall hazy image reconstruction process's computational cost and time complexity. In contrast, the LD Net method is different from the existing deep learning-based approaches. It jointly approximates the atmospheric light and the transmission map, thereby reducing the computational cost, overall complexity, time saving and the whole process can be done in a single architecture.

a) LD-Net

H. Ullah et al [14] proposed a lightweight architecture with eight convolutional layers and three concatenation layers that make up the LD-Net model. Rectified Linear Unit (ReLU) layers are placed after each convolution layer. The goal of this network is to extract multiscale characteristics from an input ($I(x)$) RGB image of size (640 x 480 x 3). This network has three concatenation layers because convolution neural networks lose low-level semantics that are learned at earlier layers when they move to higher levels of representation. Concatenation layers 1, 2, and 3 refer to these strata. Different forms of convolutional layers followed by concatenated layers are studied to keep the relevant learnt representation, and finally, an ideal layer configuration for effective reconstruction of the hazy image is chosen.

All the convolution layers present in the three-concatenation layer where Conv1 and Conv3 are combined in the Concat1, Conv4 and Conv6 are presented in the Concat2 and Concat3 layer combines the learned representations of Conv2, Conv5 and Conv7. Each concatenation layer uses a 1×1 kernel to merge the feature maps of two or more convolutional layers. The size of the image does not change in the convolution layers throughout the process. To not vary the size of the image, kernel sizes that are 1×1 , 3×3 , 5×5 and 7×7 and padding values that are 0, 1 very respective to the convolution layers and striding remains the same that is 1. The LD-Net architecture reduces the overall computational and time complexity of the framework by using no more than 16 kernels in a single convolutional layer. A given hazy image's transmission map and atmospheric light are concurrently computed. The dehazed image is given as $J(x)$ where $J(x)$ is,

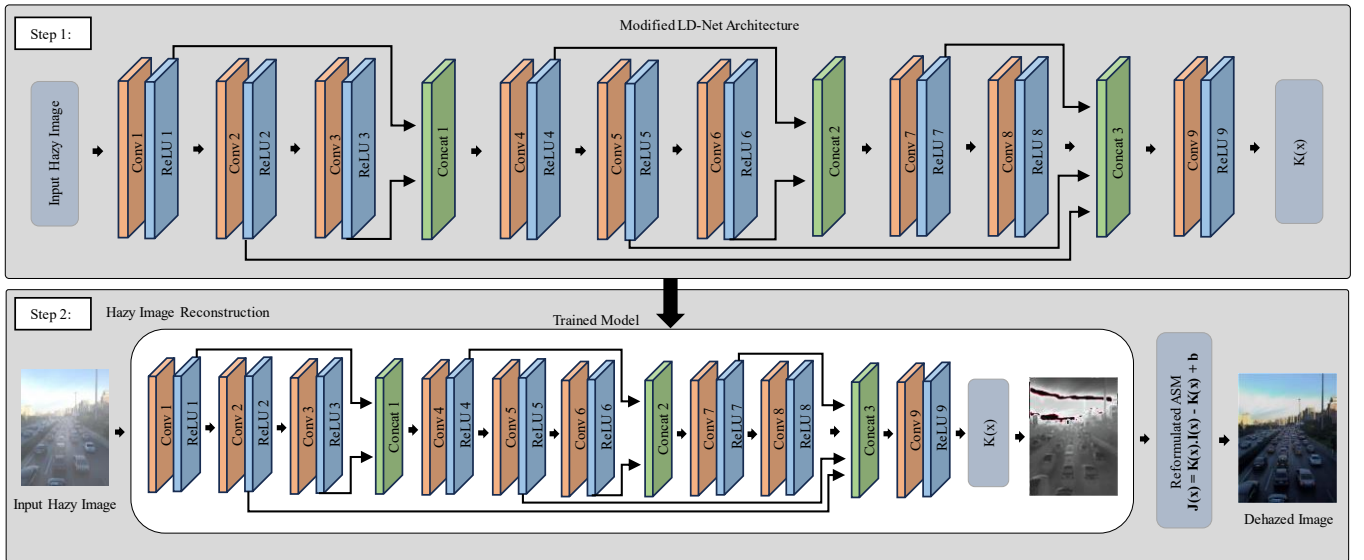


Fig. 2. There are two primary steps in the suggested model. Step 1 involves training the model, and step 2 involves passing the learned CNN model an input hazy image to recover the haze-free image

$$J(x) = K(x) \times I(x) - K(x) + b$$

Where the $K(x)$ is the output of the convolution layer holding both $t(x)$ and A , and b has a default value of 1, which is the bias term.

a) Real Time Implementation

Real-time implementation of image dehazing refers to the process of removing haze or atmospheric scattering from an image in real-time or at a speed close to real-time. This is particularly important for applications such as autonomous driving, surveillance, and real-time image enhancement, where the ability to process images quickly is critical.

To achieve real-time image dehazing, specialized hardware accelerators, such as Graphics Processing Units (GPUs) or Field-Programmable Gate Arrays (FPGAs), can be employed. These hardware platforms can significantly speed up the computation-intensive parts of the dehazing algorithms, making real-time processing feasible. Deep learning techniques, such as the Light Dehaze Net (LD-net) Convolutional Neural Networks (CNN) architecture for hazy image reconstruction, have shown considerable potential in image dehazing. Real-time implementation of deep learning-based dehazing methods can be achieved by using optimized model architectures and leveraging hardware acceleration. It can be implemented by using LD-Net on hardware boards like Jetson boards, Raspberry Pi, Zynq and Pynq.

III. MODIFIED LD-NET

We have slightly modified the LD-Net architecture [14] as shown in fig.2. Additionally, we included one extra convolution layer named Conv9. Here Conv1 and Conv3 are combined in Concat1, Conv4 and Conv6 are combined in Concat2 and the four convolution layers named as Conv2, Conv5, Conv7 and Conv8 are learned represented layers that combined in Concat3. Table-1 displays the modified LD-Net's architectural specifics. We have given the filter size 1×1 , striding 1, padding 0 and 3 output channels for conv9 layer. For training, a dataset of 13023 artificially hazy photographs with corresponding ground truth images is employed. This dataset is referred to as "The Real Time Data Set" and contains artificial hazy images with various haze

thicknesses. These fabricated photos were produced using 373 photographs taken outdoors, each of which was altered to have a distinct level of haze. Some sample images are shown in Fig. 1. We have gotten a better result than the LD-Net [14].

IV. JETSON NANO

We have used the Jetson Nano [15] for real time implementation of LD-Net for image dehazing. Because it is more feasible compared to the Raspberry Pi due to its higher computational capabilities. The Jetson Nano is designed for running AI workloads and is capable of hardware acceleration using NVIDIA's CUDA and cuDNN libraries, making it well-suited for deep learning tasks like image dehazing. The Jetson Nano is equipped with a CUDA-capable GPU (Graphics Processing Unit), this GPU enables accelerated AI computations and neural network processing. GPUs excel in image dehazing and deep learning tasks by leveraging their ability to perform thousands of parallel computations simultaneously, accelerating the process through concurrent execution. The Jetson Nano is equipped with a quad-core ARM Cortex-A57 CPU, which handles general-purpose processing tasks. It typically has 4GB of LPDDR4 memory, which helps in handling large datasets and complex neural networks. It has a microSD card slot for primary storage, allowing to use different sizes of microSD cards to store the operating system and data. It offers various I/O options, including USB 3.0 ports, HDMI output, Gigabits Ethernet, 40 GPIO (General Purpose Input/Output) pins, and more. It also supports Wi-Fi and Bluetooth connectivity. The Jetson Nano is fully compatible with NVIDIA's deep learning software development kit (SDK) called JetPack. It includes libraries, APIs, and frameworks like CUDA, cuDNN, TensorFlow, PyTorch and others, making it easier for the developers to deploy and optimize AI applications. It contains a production module that uses NVRAM instead of SD card, which describes computer memory that can continue to store information even after the power to the memory chips has been turned off.

TABLE I

Details of Proposed LD-Net's Architecture												
Layers	Conv1	Conv2	Conv3	Concat1	Conv4	Conv5	Conv6	Concat2	Conv7	Conv8	Cocat3	Conv9
Filter size	1×1	3×3	3×3		5×5	7×7	5×5		3×3	1×1		1×1
Stride, Pad	1,0	1,1	1,1	1×1	1,2	1,3	1,2	1×1	1,1	1,0	1×1	1,0
Channels	8	8	8		16	16	16		32	32		3



Fig. 3. The first row represents input hazy images to the Jetson Nano and the second row represents corresponding output dehazed images.

The NVIDIA Jetson Nano developer kit is a compact but powerful GPU processor and allows to run multiple parallel neural networks for object recognition, segmentation, speech processing, image classification and image dehazing. All of this is contained within a user friendly platform that consumes as low as 5watts. First, we installed the OS in Jetson Nano, then created a virtual environment named env to run the LD-Net model.

The Jetpack in Jetson Nano contains preinstalled libraries. Further, we installed and updated the required libraries for LD-Net like OpenCV and PyTorch in the *env*. After the completion of all the requirements we run our model inside the created environment, and the outputs are shown in the Fig. 3.

V. DOCKER CONTAINER

Creating an environment in Jetson Nano, installing, and updating libraries consumes more time. Hence, to resolve this issue, we added a feature for more compatibility and time consuming named Docker, where the Docker is a tool that helps us to package and run applications in isolated containers. Containers are lightweight, portable, and self-contained, making them ideal for deploying applications in different environments. Docker containers are similar to virtual machines, but they are much more lightweight. Virtual machines create a complete operating system environment, while Docker containers share the host operating system. This makes Docker containers much more efficient and easier to manage.

VI. IMPLEMENTATION

Implementation of modified LD-Net model is little difficult to run in Jetson Nano directly. Since, the installation of required libraries to dehaze the image in this hardware is time consuming. For instance, a library called OpenCV took more than 2 hours to install in the created environment. So to tackle this challenge the approach of the Docker container in the Jetson Nano is very useful.

TABLE II

Quantitative Comparison with each Method on NYU2 Dataset	
Method	PSNR ($\uparrow$)
MSCNN (2016)	21.27
AOD-Net (2017)	19.69
DCPDN (2018)	23.51
GFN (2018)	24.27
EPDN (2019)	26.18
Fusion coding-based (2019)	24.85
Feature forwarding-based (2019)	24.42
MSCNN-HE (2020)	21.33
Heterogeneous adversarial nets (2020)	23.85
LD-Net (2021) [14]	25.08
Proposed Model (Ours)	27.90

Docker is the inbuilt package that comes with jetpack. So we just needed to activate it. We used a Docker container from Jetson-Inference [16]. To this container we mounted all our created LD-Net model files. Finally, we run our model inside the container, it took around 30 seconds to dehaze the image. The run time analysis is corresponding to its image size. The input hazy images and the output dehazed images are shown in the fig.3. The quantitative comparison with each method on NYU2 dataset is shown in table-2. Unlike the virtual environment, the Docker container contains all the libraries that reduces the time consume. It is portable so that we can run our LD-Net model on different systems.

VII. CONCLUSION

Real time implementation of hazy images in Jetson Nano allows them clearer and easier. Image dehazing offers several advantages, which are beneficial in various practical applications. It is particularly useful in decision-making, such as surveillance, autonomous vehicles-driving reducing the risk of accidents, avoiding delaying flights and efficient video processing. Real-time dehazing algorithms can be applied to various scenarios, such as outdoor imaging, aerial photography, underwater vision, and medical imaging.

Advances in hardware have made real-time dehazing feasible on various devices with optimized resource utilization. This means that the implementation of dehazing on Jetson Nano in real-time does not require prohibitively expensive hardware, making it accessible to a broader range of applications. Moreover, the implementation on Jetson Nano using Docker container of image dehazing brings flexibility, scalability, and ease of deployment to the process, making it a powerful and practical solution for various real-world applications. Docker containers provide a lightweight, isolated environment for running applications. This isolation ensures that the image dehazing process does not interfere with other software running on the host system. Docker containers can also be easily moved between different environments, making it simple to deploy the dehazing solution across various platforms.

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